

A QUANTITATIVE INVESTIGATION INTO THE EFFECTS OF PUPIL DIAMETER AND DEFOCUS ON CONTRAST SENSITIVITY FOR AN EXTENDED RANGE OF SPATIAL FREQUENCIES IN NATURAL AND HOMATROPINIZED EYES

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Abstract—Contrast sensitivity was measured in 12 subjects for different spatial frequencies of sinusoidal grating patterns, generated by oscilloscope, for pupil diameters 2–8 mm and for defocus of +1–4 D, following homatropine eyedrops. Changes in pupil diameter, without correction for the change in retinal illumination, had no significant effect on contrast sensitivity, except at 0.5 and 1 c deg⁻¹ when a significant reduction occurred with the 2 mm pupil. Defocus caused a large reduction in contrast sensitivity at spatial frequencies higher than the peak of the contrast sensitivity function (3 c deg⁻¹) and a smaller reduction below the peak. In both individual and group results, there was no significant effect of defocus in causing a disproportionately greater reduction in contrast sensitivity at higher spatial frequencies nor were the zero minima predicted by optical theory observed. The results were confirmed in eight subjects viewing with the natural eye, though the reduction in contrast sensitivity caused by +1 D defocus was not significant: this was attributed to the relaxation of accommodation in response to defocus. To predict the performance of the visual system, multiple regression equations were derived to incorporate terms for pupil diameter, defocus and spatial frequency. These equations reflected the lack of effect of pupil diameter, while defocus caused a 51% loss in contrast sensitivity per dioptre at higher spatial frequencies (3–38 c deg⁻¹) and a 19% reduction at low spatial frequencies.

INTRODUCTION

The quality of the retinal image depends considerably on the geometrical aberrations present in the optical media, so much so that Helmholtz (1909) remarked that the monochromatic aberrations of the eye would be unacceptable in an optical instrument of good quality (p. 189). Yet, years later, Campbell (1968) was able to conclude that the optical media of the human eye were, in fact, more than adequate for the resolving power of the retina, since most of the degradation of image quality arose in the nervous system with only a small proportion attributable to the optical media. The aberrations, to which a readable introduction is given by Price (1976), were assessed for the human eye by van Meeteren (1974) who concluded that, in white light, chromatic differences of focus constituted the most important consideration. Others have, however, rated the effects of spherical aberrations as being of equal importance to chromatic aberrations (Gubisch, 1967). More recently, a new perspective on monochromatic aberrations was presented by Howland and Howland (1976; 1977) who derived the magnitudes of the different aberrations from drawings made by their subjects of a grid pattern viewed through two crossed cylindrical lenses. This approach was continued by Walsh and Charman (1985) who photographed the retinal image and derived phase as well as amplitude relationships. The important findings was that two sets of aberrations hitherto unmeasured and tacitly assumed to be of negligible importance, *viz.* coma and cylindrical aberrations (van Meeteren, 1974) were shown to predominate at all diameters of pupil.

Pupil diameter has marked effects on image quality as shown by psychophysical studies with the Landolt C (Jenkins, 1963), the Snellen test (Tucker and Charman, 1975) and sinusoidal grating patterns (Arnulf and Dupuy, 1960; Campbell and Green, 1965; van Meeteren, 1974; Charman, 1979; and Bour, 1980). Objective measurements have been made of the line-spread function by scanning photo-electrically the faintly reflected fundal image of a line of light, from which the modulation transfer characteristics of the optical media at different spatial frequencies can be calculated (Westheimer and Campbell, 1962; Campbell and Gubisch, 1966; Charman and Jennings, 1976). With monochromatic light, in a perfectly diffraction-limited system, image contrast declines approximately linearly with spatial frequency until the cut-off frequency determined by a/λ (a is pupil diameter and λ is wavelength) is reached (Westheimer, 1964). However, for the human eye, additional marked attenuation of contrast occurs, particularly in the range of intermediate spatial frequencies, due to the effects of geometrical aberrations. These aberration effects are reduced as pupil diameter decreases until an approximately diffraction-limited system is attained at pupil diameters of 2.0–3.0 mm (Arnulf and Dupuy, 1960; Bour, 1980; Campbell and Green, 1965; Campbell and Gubisch, 1966; van Meeteren, 1974; Howland and Howland, 1977). Further reductions in pupil diameter to below the optimal value lead to increased effects of diffraction from around the edge of the pupil. Comparison between theoretical predictions of the in-focus retinal image quality, based on the effects of both aberrations and

diffraction, and actual measured data for different pupil diameters, revealed close agreement (van Meeteren, 1974).

Defocus of the retinal image is potentially a more serious source of degradation in practice and may occur due to errors of refraction or accommodation. Administration of drugs may also affect accommodation (and pupil diameter), e.g. atropine-like drugs which paralyse accommodation and dilate the pupil and anti-cholinesterases which cause accommodative spasm and constrict the pupil (Rengstorff, 1970). Even in natural viewing, changes in accommodation arise. Viewing a brightly illuminated field like the featureless blue sky leads to an accommodative effort of some 1.7 D which is referred to as empty-field or space myopia (Whiteside, 1952). Such an adoption of a resting level of accommodation has also been invoked to account for instrument myopia which may be as great as 3 D for monocular viewing or 5 D for binocular viewing (Hennessey, 1975). Accommodation also increases in twilight when the eye becomes myopic by 1.5–2.0 D (Campbell and Primrose, 1953), though there is an element of 0.5–1.5 D due to increased spherical aberration resulting from dilation of the pupil (Komen *et al.*, 1951).

The theory of the effects of defocus is relatively more complex compared with that for different pupil diameters. Defocus leads to an accelerated fall in image contrast with increasing spatial frequency as illustrated in Fig. 1 which is taken from Hopkins (1955). When the defocus reaches a certain level, corresponding to $n = 2$ in Fig. 1, the transfer function of a diffraction-limited system is predicted to cross the abscissa to give contrast reversal, with the reversal occurring at lower spatial frequencies for greater defocus. The agreement between theoretical predictions and measurements for a simple glass lens is very close (Hopkins, 1962), though for measurements of contrast sensitivity and optical quality of the human eye, however, the correspondence is not so good. Contrast reversal *per se* is not a cause for concern in that contrast is detected by the visual system irrespective of phase, but minima of zero contrast have not been reported and image contrast generally is higher than predicted (Campbell and Green, 1965; Charman, 1979). On the other hand, the measurement of line spread functions does tend to show close agreement with predicted results (Charman and Jennings, 1976).

The deleterious effects of defocus are generally described as being more severe for higher spatial frequencies with an absence of effect at very low spatial frequencies (Campbell and Green, 1965; Regan *et al.*, 1977). Qualifications must be added due to the fact that Campbell and Green's observations were made at a small (2 mm) pupil diameter and Regan *et al.*'s results were part of a larger study into multiple sclerosis and involved only a limited range of spatial frequencies with one level of defocus in the natural eye. Contrast sensitivities at low spatial frequencies are, however, attenuated with a larger pupil diameter (Green and Campbell, 1965), the use of which also gives rise to a further complication when spherical aberration is present. The refractions required to bring different spatial frequencies into focus differ, e.g. focus of high spatial frequencies may require an additional +0.8 D

compared with low spatial frequencies (Green and Campbell, 1965). The visual system is also able to tolerate a degree of defocus blur. Hopkins (1962) calculated that, for a diffraction-limited system, image contrast deteriorated rapidly when the Rayleigh criterion of $\lambda/4$ (corresponding to approximately 0.03 D for a 6 mm pupil and light of wavelength 555 nm) was exceeded, whilst the psychophysical measurements of tolerance of defocus blur (Whiteside, 1957; Campbell and Westheimer, 1958) indicate a larger value of 0.20–0.25 D.

Of particular interest to us was to make measurements which could be used practically to assess the effects of defocus and pupil diameter under natural conditions in circumstances like, for example, flying a high speed aircraft at low altitude. Several reasons exist why this cannot be done from existing data. Perhaps most important is the relatively small number of subjects studied in any depth—two by Campbell and Green (1965) and one by Charman (1979). Howland and Howland (1977) who studied a total of 55 eyes reported a four-fold range of variability in contrast modulation at 20 c deg⁻¹, thus emphasizing the need for a relatively large sample. We, therefore, carried out our measurements on a total of 12 subjects in order to go some way to meeting this point. Previous studies also used cycloplegic drugs to remove the fluctuations in accommodation (Campbell *et al.*, 1959) and pupil diameter in order that these might be controlled precisely experimentally. This may of its own account adversely affect contrast sensitivity measurements and should the cycloplegic drug diffuse into the retina, neuronal activity may also be affected. Therefore, we took the precaution of repeating our experiments in a number of subjects without homatropine to check against such possible effects. In our experiments, pupil diameter was studied in the absence of compensation for changes in retinal illumination, as usually practised, to reproduce what occurs in natural viewing conditions. Likewise, defocus was studied at small pupil diameters to reflect the well-known link between accommodation and pupil diameter. Our method of assessment was to measure contrast sensitivity to sinusoidally modulated grating patterns generated on a cathode ray tube (CRT), since this provides a more complete description of visual function. This supposition is supported by the studies of Ginsburg (1981) and Ginsburg *et al.* (1982) who found a correlation between an aeroplane pilot's ability to detect a target and contrast sensitivity but not Snellen acuity. A preliminary report of our results has already been made (Kay and Morrison, 1985).

METHODS

The apparatus and experimental methods are those described in Morrison and McGrath (1985) but involving a wider range of spatial frequencies.

CRT display

Sinusoidal grating patterns were generated on a Telequipment DM53S oscilloscope with P31 phosphor, peak emission at 520 nm. The luminance of the display as measured with a Pentax Spotmeter V was 1.4 cd m⁻², being 3.8 log units above foveal threshold: both rod and cone pathways may thus be considered to be oper-

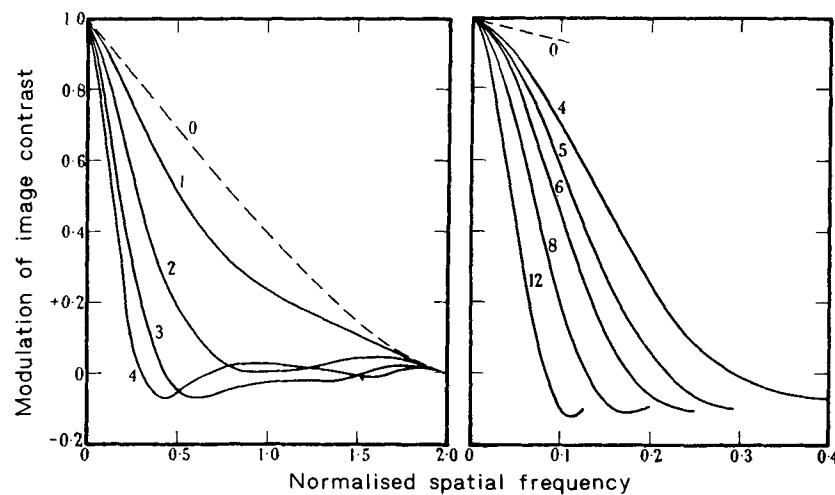


Fig. 1. Computations by Hopkins (1955) of the effects of defocus on the modulation transfer function for a diffraction-limited optical system. Spatial frequency is normalized so that the value of 2.0 corresponds to the cut-off value for the system [this cut-off is given by $(1.746 \times 10^4 a) \lambda \text{ c deg}^{-1}$ for the eye, where a is the pupil diameter in mm and λ is the wavelength in nm]. The figures on the curves give the defocus in terms of the parameter n , where the corresponding wavefront aberration is $w_{20} = (n/\pi)\lambda$. A value of $n = 1$ corresponds to a defocus of approximately 0.125 D when the eye has a pupil diameter of 3.3 mm and the wavelength is 555 nm.

Note the expanded abscissa on the right hand graph. Reproduced with permission from Hopkins (1955).

ational. The screen itself was surrounded by green card to give an aperture of width 1.5 deg when viewed at 3 m and 4.5 deg at 1 m. Two viewing distances were used in order to obtain a full range of spatial frequencies, but avoid the difficulty of a small number of cycles which would limit sensitivity. Viewing was under subdued lighting of ~ 0.5 lux.

Prior to the experiments, each subject underwent the Snellen test. If necessary, additional lenses to the nearest 0.12 D were worn in a trial frame for optimal acuity which was always 6/5 or 6/4. Contrast sensitivity was measured for one eye by an ascending method in which the experimenter steadily increased the contrast until the subject was just able to see the gratings in the centre of the screen and no more. Prior to the study proper, each subject underwent a familiarization run consisting of spatial frequencies 10, 20 and 30 c deg^{-1} . Our experience was that once a subject had successfully practised the highest spatial frequency, he/she became attuned to the method and was reasonably consistent in judgements thereafter. Consistency was actually tested in a series of control experiments described later.

Experimental protocol

The same standard protocol was followed by all subjects. Five measurements of contrast threshold were made at each of 10, 20, 30, 38, 35, 25, 15 and 10 c deg^{-1} viewed at 3 m and 8, 3, 1, 0.5 and 5 c deg^{-1} at 1 m. These were collected for the following conditions.

- (1) Natural pupil diameter and optimal accommodation.
- (2) Viewing with a 3.0 mm artificial pupil and optimal accommodation.
- (3) 2.0 mm artificial pupil and optimal accommodation.
- (4) 2.0 mm artificial pupil and +2.0 D additional defocus.
- (5) 3.0 mm artificial pupil and +2.0 D additional defocus.

- (6) 3.0 mm artificial pupil and +4.0 D additional defocus.

- (7) 3.0 mm artificial pupil and +1.0 D additional defocus.

- (8) Repeat measurement at 10, 20 and 30 c deg^{-1} with natural pupil and optimal accommodation.

The artificial pupil was worn in a trial frame which also carried the additional lenses when these were required.

Alternatively, one drop of 2% homatropine hydrobromide solution was instilled, which maximally dilated the pupil and paralysed accommodation after 20–30 min. With the aid of a purpose-made Snellen chart, the subject was optimally refracted to within 0.12 D for viewing at 1 m and 3 m, in accordance with the comments of Charman (1979) on the importance of accurate refraction. Then experiments (3)–(8) were repeated, except that (8) was with the dilated pupil. Experimental time was some 6 h with an additional 1–1½ h allowed for breaks.

Control experiments

Since our experiments were of a demanding nature, though no undue discomfort was ever reported during them, and since we could find no reports in the literature of the consistency of contrast sensitivity measurements bar Ginsburg and Cannon's (1983) study on different methods, we decided to undertake a set of control measurements. Contrast thresholds were measured at 10, 20, 30, 25, 15, 5, 3, 1 and 8 c deg^{-1} to the CRT display at four one hourly intervals and once more on a second day. These experiments were undertaken by 13 subjects, four of whom also took part in the main series of experiment.

Data analysis

t-tests and calculations of best-fitting linear regression equations were made with Minitab (Ryan, Joiner and Ryan, 1985) on the University Mainframe computer.

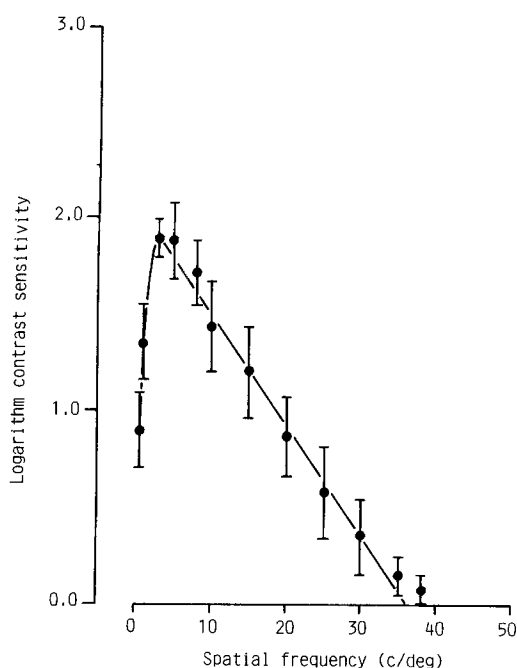


Fig. 2. Contrast sensitivities (on logarithmic scale) to CRT showing the geometric mean $\pm$ SD for 12 observers for viewing with natural 6 mm pupil.

Further comparisons of the slopes and intercepts of regression equations were made according to Draper and Smith (1981).

RESULTS

Contrast sensitivity was measured after homatropine eyedrops in 12 subjects and in the natural eye in eight subjects. The former consisted of 10 males and two females, ages 18–40 years: no differences were observed between male and female subjects nor was there an age-related trend in the group as a whole.

Contrast sensitivity to CRT display

The contrast sensitivity function (CSF) for viewing with the natural pupil of some 6 mm diameter consisted of a peak of mean value 81 units (range 62–182) at 3–5 c deg⁻¹ with fall-offs at lower and higher spatial frequencies. The high frequency fall-off was best fitted by a logarithmic-linear plot as illustrated in Fig. 2, with a cut-off frequency of typically 35–38 c deg⁻¹. For each subject the CSF was of the same form as that in Fig. 2, though there was variation in the absolute contrast sensitivity values as shown by the standard deviation bars. For any one individual the spread of individual data points was within ± 0.1 log units of the mean (Morrison and McGrath, 1985).

Homatropine eyedrops

These caused the pupil to dilate from its typical natural diameter of 6 mm under our illumination conditions to 8 mm. The additional refraction required for optimal vision was +0.25 D at 3 m and +0.75 D at 1 m, i.e. somewhat less than the expected values of +0.33 D and +1.0 D, respectively. Comparison of Figs 2 (natural pupil) and 3A (homatropine) indicates that homatropine eyedrops were without effect when the

additional lenses were being worn. This is also apparent from the regression equations for 3–38 c deg⁻¹ presented in Table 1; there was no significant difference between the intercepts or between the slopes for the two sets of data ($P > 0.25$). When paired *t*-tests were made for the highest spatial frequency detected, no significant difference was observed ($P > 0.25$). Paired *t*-tests before and after homatropine for 0.5, 1.0 and 3 c deg⁻¹ also failed to reveal a significant difference ($P > 0.25$).

Pupil diameter

In the homatropinized eye, viewing with the 3 mm artificial pupil tended to improve contrast sensitivity whilst the 2 mm pupil tended to worsen it, compared with the fully dilated pupil (Fig. 3A). Over the range 3–38 c deg⁻¹, the intercepts and slopes of the regression equations for 3 mm and 2 mm pupil were, however, not significantly different from those for the dilated pupil ($P > 0.25$ in each case). This lack of significance is also apparent from the regression equations listed in Table 1. At 0.5 and 1.0 c deg⁻¹, the paired *t*-test showed that contrast sensitivities for the 3 mm pupil were not significantly different from those for the dilated pupil ($P > 0.25$); however those for the 2 mm pupil were significantly reduced ($P < 0.01$, in each case).

In the natural eye, much the same results were obtained as those shown in Fig. 3A, viz. the 3 mm pupil improved marginally and the 2 mm worsened marginally contrast sensitivity compared with the natural 6 mm pupil (Fig. 3B). The regression equations for 3–38 c deg⁻¹ had intercepts and slopes which were not significantly different ($P > 0.25$). Contrast sensitivity was unaffected at 0.5 c deg⁻¹ with 2 mm and 3 mm pupils (paired *t*-test, $P > 0.25$) but a significant reduction did occur with 2 mm and 3 mm pupils at 1 c deg⁻¹ ($P < 0.05$). It is therefore concluded that, under our conditions, changes in pupil diameter without compensatory changes in retinal illumination have no significant effect on contrast sensitivity in both natural and homatropinized eyes.

Defocus

Increasing defocus by positive spherical lenses with the 3 mm artificial pupil had an increasingly deleterious effect on contrast sensitivity in homatropinized (Fig. 4A) and natural eyes (Fig. 4B). In both cases the impression is of a step-wise downward shift of the 3–38 c deg⁻¹ segment of the CSF, with a smaller decrease below 3 c deg⁻¹. Essentially similar results were also obtained for the 2 mm pupil with optimal focus and with the +2.0 D lens. The regression equations for 3–38 c deg⁻¹ are presented in Table 2. For increasing defocus, the very marked reductions in the intercepts of the regression equations (Table 2) were significantly different when compared with optimal focus in homatropinized and natural eyes ($P < 0.001$). The only exception was with +1.0 D lens in the natural eye when the apparent reduction in intercept was not significantly different ($P > 0.25$).

Comparison of the slopes of the regression lines shows an apparent increase in negative slope with increasing defocus, indicating that higher frequencies were more severely affected (Table 2). However, comparison of the regression equation for each defocus with

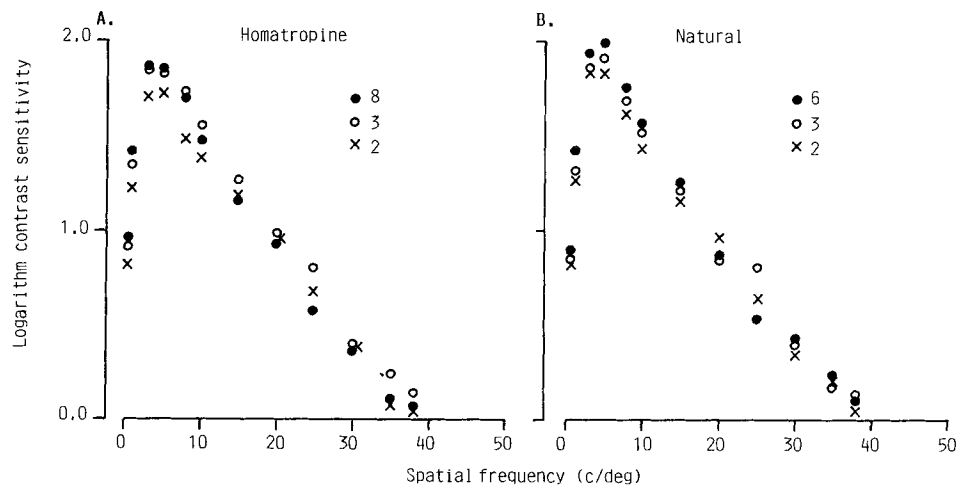


Fig. 3. Effects of pupil diameter on mean logarithmic contrast sensitivities in the homatropinized (A) and natural eye (B). Pupil diameter in mm is indicated.

Table 1. Effect of pupil diameter on the regression line of mean log contrast sensitivity (y) against spatial frequency (x) in the range 3–38 c deg⁻¹

Condition	Regression	SD of y	r	P
Pre-homatropine	$y = 2.08 - 0.056x$	0.08	-0.992	< 0.001
Homatropine/8 mm	$y = 2.07 - 0.058x$	0.05	-0.997	< 0.001
Homatropine/3 mm	$y = 2.06 - 0.052x$	0.06	-0.996	< 0.001
Homatropine/2 mm	$y = 1.94 - 0.052x$	0.05	-0.996	< 0.001
Natural/6 mm	$y = 2.15 - 0.057x$	0.12	-0.987	< 0.001
Natural/3 mm	$y = 2.07 - 0.053x$	0.08	-0.992	< 0.001
Natural/2 mm	$y = 2.00 - 0.052x$	0.05	-0.996	< 0.001

y is logarithm contrast sensitivity and x is spatial frequency.

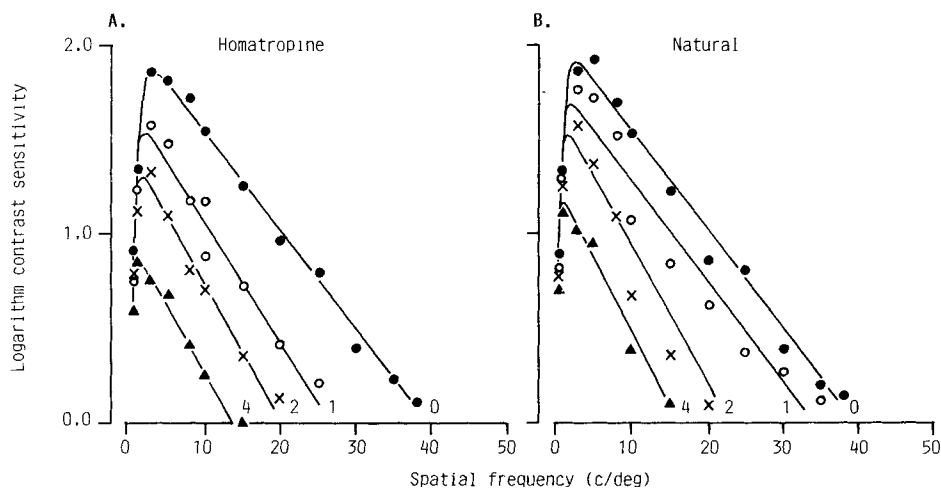


Fig. 4. Effects of positive defocusing lenses on mean logarithmic contrast sensitivities for viewing with the 3 mm artificial pupil in homatropinized (A) and natural (B) eyes. Defocus in dioptres is given alongside the appropriate contrast sensitivity curves.

that for optimal focus showed the slopes not to be significantly different ($P > 0.25$, in each case). The same absence of a significant difference between slopes was also present in each one of the individual results which are listed in Table 3.

In the homatropinized eye, where defocus was precisely controlled, the subjects fell into groups. Subjects 1 and 2 (who happened to be the authors) had slopes which were essentially unchanged with defocus.

At the opposite end of the spectrum, subjects 10–12 had slopes which increased stepwise as defocus increased in the same way as Campbell and Green's (1965) results, shown as "C and G" (a slope value of $m = -0.046$ was also obtained for +0.5 D). For subjects 3–8, the trend was for the slope to increase with +1.0 D and then to remain constant or decrease with greater defocus. Viewing with the natural eye was not without difficulties since continual shifts of focus were reported, presum-

Table 2. Effect of defocus with positive lenses on the regression line of mean log contrast sensitivity (ν) against spatial frequency (x) in the range 3–38 c deg⁻¹ for the combinations of defocusing lens power and artificial pupil diameter indicated

Condition	Regression	SD of y	r	P
After homatropine:				
0.0 D/3 mm	$y = 2.06 - 0.052x$	0.06	-0.996	< 0.001
+1.0 D/3 mm	$y = 1.69 - 0.063x$	0.10	-0.980	< 0.001
+2.0 D/3 mm	$y = 1.44 - 0.068x$	0.07	-0.986	< 0.001
+4.0 D/3 mm	$y = 0.96 - 0.068x$	0.04	-0.986	< 0.01
0.0 D/2 mm	$y = 1.94 - 0.052x$	0.05	-0.996	< 0.001
+2.0 D/2 mm	$y = 1.56 - 0.076x$	0.10	-0.979	< 0.001
Natural eye:				
0.0 D/3 mm	$y = 2.07 - 0.053x$	0.08	-0.992	< 0.001
+1.0 D/3 mm	$y = 1.79 - 0.053x$	0.15	-0.968	< 0.001
+2.0 D/3 mm	$y = 1.64 - 0.072x$	0.19	-0.950	< 0.001
+4.0 D/3 mm	$y = 1.26 - 0.077x$	0.07	-0.982	< 0.001
0.0 D/2 mm	$y = 2.00 - 0.052x$	0.05	-0.996	< 0.001
+2.0 D/2 mm	$y = 1.71 - 0.072x$	0.18	-0.956	< 0.001

Table 3. Slopes of regression lines of log contrast sensitivity against spatial frequency (c deg⁻¹) for 12 individual subjects with a 3 mm artificial pupil under the various conditions of defocus indicated. Also given are the slope values found by Campbell and Green (1965) and Regan *et al.* (1977)

	Homatropine				Natural			
	0.0 D	+1.0 D	+2.0 D	+4.0 D	0.0 D	+1.0 D	+2.0 D	+4.0 D
1. CDK	-0.054	-0.058	-0.059	-0.054	-0.063	-0.078	-0.095	-0.054
2. JDM	-0.049	-0.044	-0.042	-0.050	-0.063	-0.060	-0.048	-0.069
3. GW	-0.050	-0.061	-0.058	-0.067	-0.051	-0.052	-0.074	-0.124
4. AR	-0.048	-0.066	-0.058	-0.066	-0.048	-0.048	-0.069	-0.072
5. RO	-0.070	-0.085	-0.090	-0.083	-0.060	-0.075	-0.083	-0.077
6. JMI	-0.050	-0.061	-0.060	-0.059	-0.043	-0.069	-0.088	-0.076
7. HMD	-0.052	-0.073	-0.067	-0.059				
8. RR	-0.053	-0.074	-0.088	-0.080	-0.060	-0.067	-0.093	-0.069
9. DQ	-0.060	-0.043	-0.059	-0.081				
10. SON	-0.045	-0.059	-0.071	-0.085	-0.064	-0.068	-0.082	-0.075
11. LCC	-0.050	-0.058	-0.064	-0.073				
12. VK	-0.047	-0.055	-0.076	-0.093				
C and G	-0.048	-0.056	-0.060					
R, S and M					-0.064	-0.093		

Units: log units per cycle per degree.

ably as the accommodation mechanism hunted vainly for optimal focus. In general, the variability of the data was appreciably greater than in the homatropinized eye as shown in Fig. 4B and Table 2. Subject 2 again had reasonably constant slopes with increasing defocus. Subjects 3 and 4 showed a monotonic increase while subjects 1, 5, 6, 8 and 10 were characterized by an increase followed by a fall in slope at +4.0 D. The data of Regan *et al.* (1977) shown as ‘‘R, S and M’’ indicate an increase in slope for +1.0 D. For the data of Campbell and Green and Regan *et al.*, and in Table 3, none of the regression lines with defocus were significantly different from optimal focus ($P > 0.25$ in every case).

Increased defocus in the homatropinized eye caused a significant reduction in contrast sensitivity at 0.5 and 1.0 c deg⁻¹ with the 3 mm pupil ($P < 0.02$) but not with the 2 mm pupil ($P > 0.1$). In both 2 mm and 3 mm pupils, contrast sensitivity at 3 c deg⁻¹ was significantly reduced with defocus ($P < 0.001$). Slight differences occurred with the natural eye: at 0.5 and 1.0 c deg⁻¹, +1.0 D defocus did not cause a significant reduction ($P > 0.1$), i.e. similar to 3–38 c deg⁻¹. With greater

defocus, contrast sensitivity was significantly reduced with the 3 mm pupil ($P < 0.02$) but not with the 2 mm pupil ($P > 0.25$).

Control experiments

The results of systematically measuring CRT contrast sensitivities over 1–30 c deg⁻¹ at four one hourly intervals and on a second day are shown in Fig. 5. The significance levels of paired *t*-tests between a given trial and the first trial are shown. On day 1, there were no significant changes in contrast sensitivity except for the third and fourth trials at 1, 5 and 30 c deg⁻¹ when a significant improvement occurred. Taking the CSF as a whole from 3–30 c deg⁻¹, there were no significant changes between intercept and slope between the first and subsequent trials ($P > 0.25$). Comparisons between the fourth trial on day 1 and the trial on day 2 showed a significant improvement on day 2 at all spatial frequencies except 1 and 30 c deg⁻¹ (Fig. 5). These results were in keeping with comparisons for 10, 20 and 30 c deg⁻¹ at the start and end of the experimental session.

Despite the arduous session endured by everyone, the

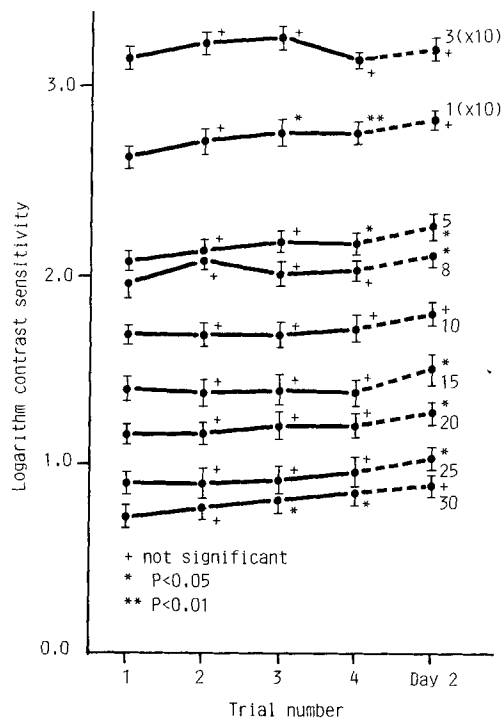


Fig. 5. Reproducibility of contrast sensitivity measurements. Mean $\pm$ SE for 13 observers at spatial frequencies stated (c deg^{-1}) and four consecutive hourly trials. Significance levels refer to comparisons by paired t -test between a given trial and trial 1. Day 2 is compared with trial 4.

tendency was for the repeat readings to be the same or slightly higher than those at the start of the session. Thus, contrast sensitivities measured on the same day were consistent, though this was not necessarily true for different days.

DISCUSSION

In the present study, the effects of changes in pupil diameter and defocus on contrast sensitivity have been investigated. The new aspects of this study are that a relatively large sample of subjects was tested and that viewing was done with the natural eye as well as the homotropinized eye. Changes in pupil diameter were made without compensation for changes in retinal illumination, such as occurs in real life, and a relatively wide range of defocuses was studied. By repeating the measurements of contrast sensitivity for the full range of spatial frequencies, reproducibility was found to be good, though this would not necessarily have been the case if the measurements had been spread over several days. The interpretation of contrast sensitivities at low spatial frequencies does, however, require some discussion. Spatial frequencies at 8 c deg^{-1} and below were tested from the near distance of 1 m where the display subtended 4.5 deg . It has been demonstrated by McCann *et al.* (1974) that the viewing of low spatial frequencies is unaffected by distance and the results from near and far distances may reasonably be combined (Campbell and Robson, 1968). Two other considerations are, however, of importance. First, contrast sensitivity depends upon the number of grating cycles visible as well as spatial frequency. According to

Hoekstra *et al.* (1974), the minimum number of cycles at the screen luminance used in our experiment is 3 cycles. At 1 c deg^{-1} when 4 cycles were visible, contrast sensitivity may be expected to be normal, whereas at 0.5 c deg^{-1} when only 2 cycles were visible, there would be this additional factor of a reduced number of cycles causing a further reduction in contrast sensitivity. This factor, however, would be present for all the subjects tested. Second, contrast sensitivity measurements are additionally depressed when the surround is darker than the CRT display, as was the case in our experiments (Estevez and Cavanus, 1976). While this effect would be present for all subjects, it would not be constant throughout our experiments since the effect of defocusing lenses would be to reduce the luminance gradient from CRT to surround. The consequence of this would be to tend to reduce the depressant effect of the defocusing lenses observed in the present study. This factor would, however, be present for all subjects and for viewing an object against a dark background, our results would still be directly relevant.

Pupil diameter

Changes in pupil diameter were in general without significant effect on contrast sensitivity. Given the substantial body of evidence that reduction of the pupil diameter *per se* improves visual resolution (Jenkins, 1963) and optical quality (Arnulf and Dupuy, 1960; Campbell and Green, 1965; Campbell and Gubisch, 1966; Gubisch, 1967; van Meeteren, 1974; Howland and Howland, 1977; Charman, 1979; Bour, 1980), our result is probably most readily explained by this improvement being annulled by the accompanying reduction in retinal illumination. A similar proposal has been made in respect of the effects of small pupil diameters on visual resolution at low screen luminances (Woodhouse, 1975). Our results also confirm those of a previous study in which the wearing of a 3 mm pupil was without effect on contrast sensitivity in the natural eye (Morrison and McGrath, 1985). The significant reduction in contrast sensitivity by the 2 mm pupil at 0.5 and 1.0 c deg^{-1} while the 3 mm pupil was without effect is anomalous. If one seeks to explain this on the basis of reduced retinal illumination, all the evidence indicates that reduced illumination tends to spare the lowest spatial frequencies (Schade, 1956; DePalma and Lowry, 1962; van Meeteren and Vos, 1972). Possibly the increased effects of diffraction at 2 mm pupil diameter were sufficient, especially when coupled with reduced retinal illumination, to degrade contrast sensitivity. This is supported by the tendency for contrast sensitivities over $3\text{--}38 \text{ c deg}^{-1}$ to be reduced at 2 mm compared with 3 mm pupil diameter, although the reduction was not statistically significant.

Defocus

This had a profound effect, causing a stepwise reduction in contrast sensitivity with increasing defocus. A number of deviations from predicted theory (Hopkins, 1955) and from the study of Campbell and Green (1965) was, however, apparent. The lowest spatial frequencies were not unaffected though they were reduced less than higher spatial frequencies (Figs 4A and

4B) while the highest spatial frequencies were not disproportionately affected compared with lower spatial frequencies. The discrepancies with Campbell and Green's work are, however, fairly readily explained. Their study, in which low spatial frequencies were not attenuated by defocus, was made with a 2 mm artificial pupil. With a 7 mm pupil diameter low spatial frequencies were markedly reduced (Green and Campbell, 1965) and there is an indication that this was also the case with a 4 mm pupil at large defoci (Charman, 1979). Our results showed a significant reduction with the 3 mm but not with the 2 mm pupil. The explanation may lie in the depth of focus which would be greater with the 2 mm pupil, especially when coupled with the accompanying reduction in retinal illumination which would also increase the depth of focus (Campbell, 1954), so resulting in a smaller effect of defocus on contrast sensitivity. While a tendency for higher spatial frequencies to have a greater roll-off of contrast sensitivity with defocus was noted in our results, it was never statistically significant on an individual basis or group basis. Campbell and Green had based their conclusion that there was a greater effect on high spatial frequencies from their plot of contrast sensitivity against defocus for a selected range of spatial frequencies. Inspection of this figure (their Fig. 6) shows that it depends very much on the precision of the data points for 30 c deg⁻¹ for large defocus. If instead the full CSFs for different defoci (their Fig. 7) are analysed statistically, the regression equations for different defoci are not significantly different from optimal focus ($P > 0.25$, in each case). Further, with 7 mm diameter pupil, the downward shift is clearly parallel (Green and Campbell, 1965) as is the case with Charman's (1979) data over 5–30 c deg⁻¹ for different pupil diameters.

In the homotropinized eye, three out of our 12 subjects had an increased roll-off with defocus (though this was not statistically significant), two had constant slopes while the majority showed an interesting pattern (Table 3). An increase in slope occurred between optimal focus and +1.0 D but above this, the slope stayed constant or even decreased. This is contrary to the notion of an increased effect of defocus on higher spatial frequencies which should be a continuous function of defocus. Two explanations for the effect of the +1.0 D lens in increasing slope are possible. The simpler is that it is merely the consequence of placing a glass lens before the eye with the result that additional lenses would be without effect on the slope, which was often the case. While we omitted to exclude this possibility by the use of zero power glass lenses, in another context, *viz.* depth judgements, defocus caused by glass lenses was similar in effect to changes in accommodation caused by a split mirror apparatus by which apparent movement of the target was effected (Morrison and Whiteside, 1984). The second explanation concerns the difference in optimal focus for low and high spatial frequencies, which is particularly apparent at large pupil diameters. At 6–7 mm pupil diameter, the difference in optimal focus between high and low spatial frequencies is –0.8 D (Green and Campbell, 1965). While this difference disappears at 2 mm pupil diameter, presumably due to reduction of geometrical aberrations (Charman

and Heron, 1979), some difference may be present with the 3 mm pupil diameter used in our study. The entire depth of focus at 3 mm pupil diameter and 3 cd m⁻² luminance, which is some +0.6 D (Campbell, 1958), would be available to offset a defocus error since accommodation is located at the hyperopic point of the depth of focus rather than at the midpoint of the range (Kasai *et al.*, 1971). Hence a residual difference of some –0.2 D may be present at low spatial frequencies which would partially compensate for the external lens at lower spatial frequencies leading to an increase in slope of the contrast sensitivity function. For greater defocus, low and high spatial frequencies would then be defocused to the same degree, resulting in unchanged slopes of the CSF. The conclusion to be drawn is that defocus does not have a greater effect at higher spatial frequencies.

The predicted zero minima which, with defocus of 1.0 D and a pupil diameter of 4 mm, occur at about 6.6 and 10 c deg⁻¹ in the diffraction-limited eye (Smith, 1982) were also not apparent in the present results nor in those of Campbell and Green (1965). The results of Charman (1979) did show undulations but these were not always coincident with the theoretical minima. Minima are apparent in the results of Bour (1980) though the detection of the grating pattern was done only for 56% and 100% contrast. Nevertheless, given that all the studies employed artificial pupils which ought to have been small enough to make the eye almost free of monochromatic aberration, the deviation from theoretical predictions indicates that the optical media of the eye are more complex than a simple diffraction-limited system.

In the present study, similar results were obtained with the natural eye as with the homotropinized eye with certain exceptions. It was notable that, during the experiments, accommodation did not remain constant with the defocusing lens but fluctuated noticeably, presumably as the visual system sought sharp focus. Then another factor adding to the variability is that different individuals respond differently to positive lenses: some relax what accommodation they have, many do not change their level of accommodation at all, while some actually increase their accommodation (Reese and Fry, 1941). Generally, the reduction in contrast sensitivity caused by the +1 D lens was not statistically significant and, in two out of our eight subjects who viewed with natural eyes, contrast sensitivities to +1 D defocus were indistinguishable from optimal focus, indicating that these belonged to the category of individuals who relaxed their accommodation to the external lens. Hence relaxation of some 0.25 D for viewing at 3 m and 0.75 D at 1 m, when added to a depth of focus of 0.6 D with the 3 mm pupil (Campbell, 1958) and a tolerance to defocus blur of 0.20–0.25 D (Whiteside, 1957; Campbell and Westheimer, 1958), would be sufficient to compensate for the external +1 D lens. Since the latter two factors were also present in the homotropinized eye, when the +1.0 D lens did significantly reduce contrast sensitivity, this highlights the marked effect that even a small degree of defocus might have on the CSF. This is consistent with the prediction of Hopkins (1955) that the optical transfer function will be markedly depressed for defocus of greater than ~ 0.125 D (corresponding to $n = 1$ in

Fig. 1, for a pupil diameter of 3 mm and green light).

Predictions of contrast sensitivity

In order that our measurements might be of practical value in predicting contrast sensitivity at our CRT luminance, we derived the best-fitting multiple regression equations in terms of pupil diameter, defocus and spatial frequency. To simplify the analysis, the CSF was separated at the peak of 3 c deg⁻¹ into low and high spatial frequency limbs and analysed in both natural and homatropinized eyes. The most appropriate equations were:

Homatropine

$$0.5-3 \text{ c deg}^{-1}, y = 0.80 + 0.29x_F + 0.021x_P - 0.09x_D, \\ SD_y = 0.15, r = +0.924 (P < 0.001)$$

$$3-38 \text{ c deg}^{-1}, y = 1.99 - 0.055x_F + 0.009x_P - 0.31x_D, \\ SD_y = 0.11, r = -0.981 (P < 0.001).$$

Natural

$$0.5-3 \text{ c deg}^{-1}, y = 0.78 + 0.33x_F + 0.024x_P - 0.05x_D, \\ SD_y = 0.15, r = +0.926 (P < 0.001)$$

$$3-38 \text{ c deg}^{-1}, y = 2.05 - 0.057x_F + 0.016x_P - 0.26x_D, \\ SD_y = 0.15, r = -0.972 (P < 0.001)$$

where y is logarithm contrast sensitivity, x_F is spatial frequency (c deg⁻¹), x_P is pupil diameter (mm) and x_D is defocus (dioptries).

Further analysis at 0.5–1.0 c deg⁻¹ in the homatropinized eye gave coefficients for pupil diameter and defocus of +0.017 and –0.08, respectively, i.e. similar to those above, indicating that the inclusion of 3 c deg⁻¹ in the above equations had not biased these terms. In general, pupil diameter had only a very small effect on contrast sensitivity. On the other hand, the effect of defocus was profound, particularly over 3–38 c deg⁻¹, causing reductions of 51% and 45% per dioptre in homatropinized and natural eyes, respectively. At 0.5–3 c deg⁻¹, the reductions were smaller, viz. 19% and 11% per dioptre, respectively. This relative sparing of low spatial frequencies has important consequences. Object detection should be relatively unimpaired since this has been correlated with the peak of the contrast sensitivity function rather than the highest spatial frequency (Ginsburg, 1981; Ginsburg *et al.*, 1982). Stereopsis should also be relatively unimpaired, in that fusion of the two images can occur with low spatial frequencies even in the presence of conflicting high spatial frequency information (Julesz, 1971). Stereoacuity will, however, be degraded to much greater degree than visual acuity since a full range of spatial frequencies is required for this task (Westheimer and McKee, 1980). Initiation of the accommodative mechanism should also be unaffected but the attainment of sharp focus which depends upon high spatial frequencies (Charman and Heron, 1979) would be impaired. The smaller effect of defocus in the natural eye probably reflects the capacity to relax accommodation in response to defocus.

The deleterious effect of defocus in our study is somewhat smaller than the 65% per dioptre reduction calculated from Campbell and Green's Fig. 7 and the 77% per dioptre from Regan *et al.*'s (1977) data. Both sets of investigators used a much higher screen luminance and appear to have used a descending method to determine contrast threshold. In inexperienced subjects

this may lead to underestimation of contrast threshold (Ginsburg and Cannon, 1983) and it is noteworthy that Regan *et al.*'s subjects had very high values of slope (Table 3) indicating that, perhaps, their subjects found greater difficulty at higher compared with lower spatial frequencies. In Campbell and Green's case, the data from only one subject were presented and thus sampling may have been a factor accounting for this difference.

In simplifying our results, the term for pupil diameter could reasonably be ignored since its effect is so small. Hence, in the mesopic range of illumination, for up to 4 dioptres of defocus, it may be expected that contrast sensitivity will be reduced by half for each dioptre of defocus, for all but the very lowest of spatial frequencies.

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